

NITHIN S. KUMAR

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PROFESSIONAL SUMMARY

My research focuses on bridging **physics-based and learned models** to enable stable, dynamic, and interaction-rich behavior in **continuum robots**. Prior to my Ph.D., I spent four years in **industrial R&D** leading multidisciplinary projects spanning mechanical design, automation, and manufacturing process development.

EDUCATION

Vanderbilt University Ph.D. in Mechanical Engineering <i>Dissertation: Dynamic Modeling and Control of Soft Continuum Robots; Advisor: Prof. Eric J. Barth</i>	Nashville, TN 2023 – Expected 2026
Worcester Polytechnic Institute (WPI) M.S. in Robotics Engineering	Worcester, MA 2022
Massachusetts Institute of Technology (MIT) Mechanical Engineer's Degree, Mechanical Engineering S.M. in Mechanical Engineering <i>Thesis: Loading as a Design Parameter for Genetic Circuits; Advisor: Prof. Domitilla Del Vecchio</i>	Cambridge, MA 2018 2016
Vanderbilt University B.E. in Mechanical Engineering, GPA: 3.99/4.00	Nashville, TN 2014

HONORS & AWARDS

ASME Dynamic Systems and Control Division Rising Star Award Selected for research contributions in dynamic systems and control.	2025
Vanderbilt University School of Engineering Salutatorian	2014
Mechanical Engineering Award First-ranked graduating student in Mechanical Engineering.	2014
Vanderbilt University Full-Tuition Scholarship	2010 – 2014

PUBLICATIONS

Journal Articles

- [J7] **Nithin S. Kumar**, John E. Peters, Thomas Beckers, and Eric J. Barth. “Data-Driven Dynamic Modeling and Control of Pneumatic Soft Actuators with Stability Analysis.” *ASME Letters in Dynamic Systems and Control*, accepted, 2026.
- [J6] **Nithin S. Kumar**, John E. Peters, Sung Y. Kim, Joshua Gaston, D. Caleb Rucker, and Eric J. Barth. “Constrained Lagrangian-Based, Nonholonomic Modeling and Validation of a Pneumatically Actuated Wheeled Soft Robot.” *Journal of Mechanisms and Robotics*, 18(4), 041002, 2026. [\[DOI\]](#)
- [J5] Sung Y. Kim, **Nithin S. Kumar**, and Eric J. Barth. “Variable Stiffness and Impedance Control of a Soft Pneumatic Actuator With Tunable Collision Dynamics and Trajectory Following.” *Journal of Dynamic Systems, Measurement, and Control*, 148(3), 031003, 2026. [\[DOI\]](#)
- [J4] Joshua B. Gaston, **Nithin S. Kumar**, Eric J. Barth, and D. Caleb Rucker. “A 3D Discrete Elastic Rod Model and Observer for Continuum Robots.” *Journal of Mechanisms and Robotics*, 18(3), 031002, 2026. [\[DOI\]](#)
- [J3] Elissa D. Ledoux, **Nithin S. Kumar**, and Eric J. Barth. “Design, Modeling, and Preliminary Evaluation of a Simple Wrist-Hand Stretching Orthosis for Neurologically Impaired Patients.” *Wearable Technologies*, 5, e19, 2024. [\[DOI\]](#)
- [J2] Eric J. Barth, Mark E. Hofacker, and **Nithin S. Kumar**. “A Free-Liquid-Piston Engine Compressor for Compact Robot Power.” *ASME Dynamic Systems and Control Magazine*, 2013.
- [J1] M. T. Z. Myint, **Nithin S. Kumar**, G. L. Hornyak, and J. Dutta. “Hydrophobic/Hydrophilic Switching on Zinc Oxide Micro-Textured Surface.” *Applied Surface Science*, 264, pp. 344–348, 2012. [\[DOI\]](#)

Conference Papers

- [C11] Mohammad Ali, **Nithin S. Kumar**, Eric J. Barth, and Thomas Beckers. “Learning-Based Modeling of Soft Robots via Cosserat Rod Theory.” *IEEE Conference on Decision and Control (CDC)*, accepted, 2026.
- [C10] James C. Gallentine, **Nithin S. Kumar**, and Eric J. Barth. “Toward Frequency-Addressable Resonant Valves for High Degree-of-Freedom Hydraulic Control.” *ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, 2025. [\[DOI\]](#)
- [C9] **Nithin S. Kumar**, Jason A. Shrand, Susheela Sharma, S. Duke Herrell III, Nicholas L. Kavoussi, Ahmed Ghazi, Eric J. Barth, Ipek Oguz, Alan Kuntz, and Robert J. Webster III. “Touching is Believing: Contact Sensing to Enhance Situational Awareness for Concentric Tube Robots.” *Hamlyn Symposium on Medical Robotics*, 2025.
- [C8] Hao Li, Jiacheng Wang, **Nithin S. Kumar**, Jesse d’Almeida, Daiwei Lu, Ayberk Acar, John Han, Qingyun Yang, Tayfun Efe Ertop, Jie Ying Wu, Robert J. Webster III, and Ipek Oguz. “Automated Segmentation of Central Airway Obstruction from Endoscopic Video Stream with Deep Learning.” *SPIE Medical Imaging: Image-Guided Procedures, Robotic Interventions, and Modeling*, 2025. [\[DOI\]](#)
- [C7] James C. Gallentine, **Nithin S. Kumar**, and Eric J. Barth. “A Compact Electropermanent Magnet Valve Geometry for Switching of Magnetorheological Fluid Using Particle Jamming.” *ASME/BATH Symposium on Fluid Power and Motion Control*, 2024. [\[DOI\]](#)
- [C6] John E. Peters, Daniel Esser, Abby Grillo, **Nithin S. Kumar**, Robert J. Webster III, and Eric J. Barth. “Toward In-Scanner Transforaminal Epilepsy Interventions: A Compact MR-Safe Concentric Tube Actuation System.” *Hamlyn Symposium on Medical Robotics*, 2024.
- [C5] John E. Peters, Abby M. Grillo, Daniel S. Esser, Sarah J. Garrow, **Nithin S. Kumar**, Tyler Ball, Robert Naftel, Dario J. Englot, Joseph S. Neimat, William A. Grissom, Robert J. Webster III, and Eric J. Barth. “A Highly Compact, Multi-Material, Fluid Powered Actuation System for MRI-Guided Surgical Intervention.” *ASME/BATH Symposium on Fluid Power and Motion Control*, 2023. [\[DOI\]](#)
- [C4] M. A. Al-Radhawi, **N. S. Kumar**, E. D. Sontag, and D. Del Vecchio. “Stochastic Multistationarity in a Model of the Hematopoietic Stem Cell Differentiation Network.” *57th IEEE Conference on Decision and Control*, Miami Beach, FL, 2018. [\[DOI\]](#)
- [C3] **N. S. Kumar** and D. Del Vecchio. “Loading as a Design Parameter for Genetic Circuits.” *American Control Conference*, Boston, MA, 2016. [\[DOI\]](#)
- [C2] **N. S. Kumar**, M. E. Hofacker, and E. J. Barth. “Design of a Stirling Thermocompressor for a Pneumatically Actuated Ankle-Foot Orthosis.” *6th ASME Dynamic Systems and Control Conference*, 2013. [\[DOI\]](#)
- [C1] M. E. Hofacker, **N. S. Kumar**, and E. J. Barth. “Dynamic Simulation and Experimental Validation of a Single Stage Thermocompressor for a Pneumatic Ankle-Foot Orthosis.” *ASME/BATH Symposium on Fluid Power and Motion Control*, Sarasota, FL, 2013. [\[DOI\]](#)

Manuscripts Under Review

- [S2] **Nithin S. Kumar** and Eric J. Barth. “Control-Oriented Learning for Dynamic Tracking and Stability Analysis of Soft Pneumatic Actuators.” Under review at *IEEE Robotics and Automation Letters*, 2026.
- [S1] **Nithin S. Kumar**, Joshua Gaston, D. Caleb Rucker, and Eric J. Barth. “Task-Space Model-Based Control of Pneumatic Soft Actuators.” Under review at *IEEE/ASME Transactions on Mechatronics*, 2026.

Ongoing Research Directions

- Learning-augmented dynamic modeling and contact estimation for continuum robots, extending momentum-observer methods to learned models.
- Extension of constrained Lagrangian modeling, state estimation, and dynamic task-space control from planar to spatial continuum robots.
- Data-driven modeling and interaction-aware control of tendon-driven continuum robots under environmental contact.

FUNDED PROPOSALS

Beyond Lab Robots: Hyper-MIMO Soft Robots via Robustly Stable Data-Driven Control 2026
Vanderbilt University Scaling Success Grant, \$60,000 in postdoctoral research support. PI: Eric J. Barth. *Primary proposal writer.*

Enabling Unprecedented Control Dexterity for Soft Robots 2024–2026
Vanderbilt University Seeding Success Grant, \$120,000. PI: Eric J. Barth. *Major contributor to proposal development and writing.*

TECHNICAL SKILLS

Programming & Computing: Python, MATLAB/Simulink, C++, Git/GitHub, Linux, LaTeX

Control: Real-time control, nonlinear control, optimal control, state estimation, system identification, trajectory optimization

Modeling & Simulation: Continuum mechanics, constrained Lagrangian mechanics, EDMD/Koopman methods, Gaussian processes, FEA (COMSOL)

Robotics & Experimentation: Pneumatic systems, tendon-driven systems, motion capture, optical tracking, DAQ systems, materials testing

Robotics & Software Tools: ROS 2, MuJoCo, PyTorch, OpenCV, Teensy/Arduino

Design & Fabrication: SolidWorks, Onshape, Creo, 3D printing, machining, mechanical system design

INVITED TALKS & SEMINARS

Mechanical Engineering Graduate Seminar, Vanderbilt University	Nashville, TN 2026
Modeling, Estimation and Control Conference	Pittsburgh, PA 2025

INDUSTRY EXPERIENCE

Research & Development Engineer <i>Olympic Manufacturing Group Inc.</i>	2018 – 2022 <i>Agawam, MA</i>
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- Led design and implementation of a lean manufacturing process for heat treatment of construction fasteners at approximately 200 parts/min; developed FEA simulations, optimized process parameters experimentally, and built an automated prototype manufacturing cell, **reducing production costs by approximately 20%**.
- Developed a robotic induction case-hardening process for roofing fasteners, including custom end effectors and quenching hardware, targeting simplification of heat treatment for up to **60% of company products**.
- Led development of a drone-based photogrammetry workflow for commercial-building measurement and specification of custom roof-edge restraint systems, improving measurement safety and repeatability.
- Led multidisciplinary engineering projects spanning mechanical design, automation, manufacturing process development, and experimental validation, while **supervising two engineering technicians** supporting project execution.

PROFESSIONAL CREDENTIALS

SOLIDWORKS Advanced Parts Modeling Certification	2021
Green Belt Six Sigma Certification	2021
SOLIDWORKS Essentials Training Certification	2020
Lean Leadership Training Certification	2019
Engineer-in-Training (EIT), Commonwealth of Massachusetts	2019
COMSOL Multiphysics Intensive & AC/DC Modeling Training Certification	2019
Instron Bluehill 3 Operators Course Certification	2018

TEACHING & MENTORING

Graduate Research Mentor, M.S. Student	Vanderbilt University, 2026–Present
Teaching Assistant, Linear Control Theory	Vanderbilt University, 2026
Teaching Assistant, Measurement & Instrumentation	MIT, 2017
Teaching Assistant, Mechatronics	Vanderbilt University, 2014